

1. Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

1). $\frac{2 - x + x^2 + 4x^3}{(x + 4)^3(x^2 + 4)} =$ _____

2). $\frac{x^2 - 2x + 5}{(x^2 + 9)^2 x^2 (x - 1)} =$ _____

3). $\frac{x^3 + 2x - 1}{(x^2 + 1)^2(x + 2)} =$ _____

2. For each of the following improper integral, one may use Comparison Test to determine if they are convergent or divergent. Circle the correct answers to complete the arguments of using the comparison test.

- 1). The improper integral $\int_1^{\infty} \frac{1}{\sqrt[7]{x^4 - 0.3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{1}{\sqrt[7]{x^4 - 0.3}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^4}$, D). $\frac{1}{x^7}$, E). $\frac{1}{x^{7/4}}$, F). $\frac{1}{x^{4/7}}$

- 2). The improper integral $\int_1^{\infty} \frac{\sin^2 x}{\sqrt[4]{x^5 + 2}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{1}{\sqrt[4]{x^5 + 2}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{5/4}}$, D). $\frac{1}{x^{1/4}}$, E). $\frac{1}{x^{4/5}}$, F). $\frac{1}{x^{1/5}}$,

- 3). The improper integral $\int_1^{\infty} \frac{x^3}{\sqrt{x^8 - 0.3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x^3}{\sqrt{x^8 - 0.3}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x}$, D). $\frac{1}{x^2}$, E). $\frac{1}{\sqrt{x}}$, F). $\frac{1}{x^8}$

3. Evaluate the following integrals.

1). $\int \frac{\cos(\ln x)}{x} dx$

2). $\int x^4 e^{-3x^5} dx$

3). $\int_1^2 \frac{x^5 \sqrt{x} + 1}{x^4} dx$

4). $\int \frac{5 - 2xe^x}{x} dx$

5). $\int_1^2 \frac{e^{1/x}}{x^2} dx$

6). $\int \frac{1}{x^2 \sqrt{x^2 - 9}} dx$

7). $\int_0^{\pi/6} \sin^3 \theta \cos^2 \theta d\theta$

8). $\int \tan^3 \theta \sec^6 \theta d\theta$

9). $\int \frac{\sec^4 x}{\tan^2 x} dx$

10). $\int \arctan x dx$

11). $\int x \arcsin x dx$

12). $\int xe^{-2x} dx$

13). $\int \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$

14). $\int x^{3/2} \ln x dx$

15). $\int_1^e \frac{\ln x}{\sqrt{x}} dx$

16). $\int x \cos(3x) dx$

17). $\int e^{2x} \sin x dx$

18). $\int e^x \cos x dx$

19). $\int_0^1 \frac{1}{(3x+1)^4} dx$

20). $\int \sin^4 x dx$

21). $\int \frac{2x+4}{x^2-2x} dx$

22). $\int \frac{4x+2}{x^2-2x-8} dx$

23). $\int \frac{4x+2}{x(x^2+1)} dx$

24). $\int \frac{\sqrt{x^2-9}}{x^2} dx$

25). $\int \frac{x^2}{(4+x^2)^{3/2}} dx$

26). $\int_0^3 \frac{x^2}{36-x^2} dx$

27). $\int \frac{1}{x^3 \sqrt{x^2-1}} dx$

28). $\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16-x^2}} dx$

29). $\int x^{3/2} \ln x dx$

30). $\int_0^1 \frac{x^2}{\sqrt{x^2+4}} dx$

4. Evaluate the following improper integrals or show that they are divergent.

1). $\int_1^{\infty} \frac{(\tan^{-1} x)^2}{x^2 + 1} dx$

2). $\int_1^4 \frac{1}{(x-1)^{2/3}} dx$

3). $\int_1^{\infty} \frac{3}{(x+1)^3} dx$

4). $\int_{-2}^0 \frac{1}{(x+2)^2} dx$

5). $\int_1^{\infty} \frac{\ln x}{x} dx$

6). $\int_1^{\infty} x e^{-x^2} dx$

5. Solve the following differential equations.

1). $\frac{dy}{dx} = \frac{\sqrt{x}}{e^y}$

2). $y' = y^2 \sin x$

3). $(x^2 + 1)y' = xy$

4). $\frac{dy}{dx} = \frac{y \cos x}{1 + y^2}, \quad y(0) = 1$

5). $\frac{dy}{dt} = \frac{2t + \sec^2 t}{2y}, \quad y(0) = -5.$

6). $y' = x^6 y^{-7}, \quad y(0) = 4$

7). $(2 + x^{12}) \frac{dy}{dx} = \frac{x^{11}}{y}$

8). $\frac{dR}{dx} = 4(R^2 + 9) \quad R(0) = 3.$

6. Find the average value of the function $f(x) = \frac{2x}{(1+x^2)^2}$ on $[0, 2]$.

7. Find the average value of the function $f(x) = \cos^5 x \sin^3 x$ on $[0, \pi/2]$.

8. A rod with density $\rho(x) = 6 + \sin x$ lies on the x -axis between $x = 0$ and $x = \pi$. 1). Find the mass of the rod. 2). Find the center of the mass of the rod.

9. The linear density in a rod 8 m long is $\frac{12}{\sqrt{x+1}}$ kg/m, where x is measure in meters from one end of the rod.

1). Find the mass of the rod.

2). Find the center of mass of the rod.

3). Find the average density of the rod.

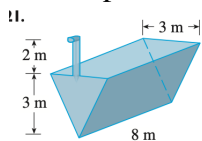
10. The force F required to stretch a spring by a distance x meters from the equilibrium (which is at $x = 0$) is given by $F = 9x$ newtons. Find the work required to stretch it 6 meters away from the equilibrium.

11. The force on a particle is described by $F(x) = 9x^3 + 3$ at a point x along the x -axis. Find the work done in moving the particle from the origin to $x = 4$.

12. Sketch the region bounded by $y = x + 2$, $y = 8 - x^2$ and $x = 0$, and then find the area of this region.

13. Sketch the region bounded by $y = 2 \sin x$, $y = 2 \cos x$, $x = 0$ and $x = \frac{\pi}{2}$, and then find the area of this region.

14. Sketch the region bounded by $y = 5 - x^2$, $y = \frac{1}{4}x^2$, and $x = 0$. and then the volume of the solid obtained by rotating this region about x -axis.
15. Sketch the region bounded $x = 1 + y^2$, $x = 0$, $y = 1$, and $y = 2$, and then find the volume of the solid obtained by rotating this region about x -axis.
16. Sketch the region bounded by $y = \frac{1}{x}$ and $y = \sqrt{x}$ on $x \in [1, 4]$. and then find the volume of the solid obtained by rotating this region about y -axis.
17. For each of the solids of revolution obtained by rotating the given region by the given rotation axis in 1)-6), complete the following Part a), b) and c).
- Sketch the graph of given region.
 - Set up integrals for computing the volume of by using both disk/washer and cylindrical shell method.
 - Choose one of the integrals you set up in Part b) to compute the volume of the solid of revolution.
- The region is bounded by $y = 4 - x^2$, $x = 0$ and $y = 0$. The solid is obtained by rotating this region about x -axis.
 - The region is bounded by $y = 4x$ and $y = \sqrt{x}$. The solid is obtained by rotating this region about x -axis.
 - The region is bounded by $y = e^x$, $x = 1$ and $y = 5$. The solid is obtained by rotation this region about y -axis.
 - The region is bounded by $x = \sqrt{y}$, $x = 0$, and $y = 4$. The solid is obtained by rotating this region about y -axis.
 - The region is bounded by $y = 2x$, $y = x^2$. The solid is obtained by rotating this region about $y = 4$.
 - The region is bounded by $y = 3x$, $y = x^2$. The solid is obtained by rotating this region about $x = 3$.
18. A particular oil tank is an upright cylinder, buried so that it is circular top is 10 meters beneath ground level. The tank has a radius of 7 meters and is 21 meter high, although the current oil level is only 15 meter deep. Find the work required to pump the oil out of the tank to the ground. The density of the oil is 800 kg/m^3 , the gravitational constant is 9.8 m/s^2 .
19. The tank of the following shape is full of water. Find the work required to pump the water out of the spout.



20. A tank in the shape of inverted circular cone has 10 meter radius and 20 meter height. It is filled full of water. Find the work required to pump all the water out of the tank to a level that is 5 meters above the top of the tank.

21. Assuming that $P \geq 0$, a population is modeled by the differential equation

$$\frac{dP}{dt} = 1.4P \left(1 - \frac{P}{4800} \right), \quad P(0) = 1200$$

- 1). For what value of P is the population increasing?
- 2). For what value of P is the population decreasing?
- 3). Write the solution of the logistic equation.
- 4). Find $\lim_{t \rightarrow \infty} P(t)$.

22. A population $P \geq 0$ obeys the logistic model. It satisfies the equation

$$\frac{dP}{dt} = \frac{1}{1300} P(13 - P), \quad P(0) = 5$$

- 1). For what value of P is the population increasing?
- 2). For what value of P is the population decreasing?
- 3). Write the solution of the logistic equation.
- 4). Find $\lim_{t \rightarrow \infty} P(t)$.

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases} \quad \int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

$$m = \int_a^b \rho(x) \, dx, \quad \bar{x} = \frac{\int_a^b x\rho(x) \, dx}{\int_a^b \rho(x) \, dx}, \quad W = \int_a^b F(x) \, dx, \quad W = \int_0^b \rho g A(y) D(y) \, dy$$

$$\text{Solutions to } \frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right), P(0) = P_0 \text{ is } P(t) = \frac{K}{1 + Ae^{-rt}}, \text{ where } A = \frac{K - P_0}{P_0}.$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined